

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Total training paths	14 · 489 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	82 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✗	✓	—	—
left thumb	✗	✓	✗	✓
left index	✗	✓	✗	✓
left middle	✗	✓	✗	✓
left ring	✗	✓	✗	✓
left pinky	✗	✓	✗	✓
left wrist	✗	—	—	—
right palm	✗	✓	—	—
right thumb	✗	✓	✗	✓
right index	✗	✓	✗	✓
right middle	✗	✓	✗	✓
right ring	✗	✓	✗	✓
right pinky	✗	✓	✗	✓
right wrist	✗	—	—	—

Enabled: 22 · Disabled: 24 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4428
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=1955 · test=98

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-* = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Pre c	W-Rec	W-F1
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Random Forest	0.9985	0.0020	0.9985	0.0020	0.9286	0.9361	0.9239	0.9186	0.9349	0.9286	0.9211
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Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_Closed_Fist	0.88	1.00	0.94	15
Double_Nothing	1.00	0.54	0.70	13
Double_Okay	0.93	0.93	0.93	14
Double_Open_Palm	1.00	1.00	1.00	14
Double_Phone	0.93	1.00	0.97	14
Double_Pistol	0.93	1.00	0.97	14
Double_Spiderman	0.88	1.00	0.93	14
accuracy			0.93	98
macro avg	0.94	0.92	0.92	98
weighted avg	0.93	0.93	0.92	98

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.9429

Per-class external accuracy

Class	Random Forest	Support
Double_Closed_Fist	1.000	5
Double_Nothing	1.000	5
Double_Okay	1.000	5
Double_Open_Palm	1.000	5
Double_Phone	0.600	5
Double_Pistol	1.000	5
Double_Spiderman	1.000	5